NETI PRELIMS REVISION SERIES

Topic 02 | Rivers of India

नेति नेति — Less noise. More clarity.

Why this matters for Prelims: Rivers appear every year in Prelims. In 2011–2019, they were mostly matching pairs — river to tributary or state. Since 2020, questions have become more specific: exact tributary sequences, left vs. right bank, rift valley origins, river tribunal details. 2024 asked the west-to-east order of rivers joining Ganga after Prayagraj. This is now a precision topic, not a general awareness one.

Quick recall trick: West-flowing peninsular rivers = "NTS-MP" — Narmada, Tapi, Sabarmati, Mahi, Periyar. Everything else flows east. Narmada and Tapi also share a second trait: both flow through rift valleys (tectonic fault zones), not over Deccan Plateau.

Brahmaputra name trail: Tsangpo (Tibet) → Dihang/Siang (Arunachal Pradesh) → Brahmaputra (Assam) → Jamuna (Bangladesh). One river, four names, one exam trap.

HIMALAYAN RIVERS (PERENNIAL — SNOW & GLACIER FED)

River	Origin	Drains Into	Key Tributaries	Unique Feature
Indus	Mansarovar/Kailas Range, Tibet	Arabian Sea (Pakistan)	L: Zaskar, Shyok, Nubra R: Jhelum, Chenab, Ravi, Beas, Sutlej	Indus Waters Treaty (1960): India keeps Ravi, Beas, Sutlej; Pakistan keeps Indus, Jhelum, Chenab
Ganga	Gangotri glacier (Bhagirathi + Alaknanda meet at Devprayag)	Bay of Bengal	L: Ramganga, Gomti, Ghaghra, Gandak, Kosi R: Yamuna, Son	India's national river (2008); largest basin (~26% of India's area)
Yamuna	Yamunotri glacier (Uttarkashi)	Joins Ganga at Prayagraj (Triveni Sangam)	R: Chambal, Betwa, Ken, Tons L: Hindon	Longest tributary of Ganga (~1376 km)
Brahmaputra	Chemayungdung glacier near Mansarovar, Tibet	Bay of Bengal (via Bangladesh)	L: Subansiri, Kameng, Manas, Sankosh R: Lohit, Dibang, Burhi Dihing	4 names: Tsangpo (Tibet) → Dihang/Siang (AP) → Brahmaputra (Assam) → Jamuna (Bangladesh). Passes through world's deepest gorge before entering India.

Kosi	Nepal Himalayas (Seven Kosi tributaries)	Joins Ganga in Bihar	Sun Kosi, Arun, Tamur (main branches)	"Sorrow of Bihar" — shifts course frequently; major cause of Bihar floods

NOTE: Left/right bank = river's left/right when flowing downstream. Ganga's most tested left-bank tributaries: Ghaghra, Gandak, Kosi. Right-bank: Yamuna, Son.

PENINSULAR RIVERS — WEST FLOWING (DRAIN INTO ARABIAN SEA)

River	Origin	Drains Into	States	Key Feature
Narmada	Amarkantak plateau, Anuppur (MP)	Arabian Sea (Gulf of Khambhat)	MP, MH, GJ	Flows through rift valley (tectonic fault). Forms estuary — not delta. Sardar Sarovar Dam on it.
Tapi (Tapti)	Multai, Satpura range, Betul (MP)	Arabian Sea (Gulf of Khambhat)	MP, MH, GJ	Also flows through rift valley — parallel to Narmada. No delta.
Sabarmati	Aravalli hills, Udaipur (Rajasthan)	Gulf of Khambhat	RJ, GJ	Ahmedabad & Gandhinagar on its banks. Gandhi Ashram (Sabarmati Ashram) here.
Mahi	Vindhya Range, Dhar (MP)	Gulf of Khambhat	MP, RJ, GJ	Crosses the Tropic of Cancer twice — unique feature UPSC has tested.
Luni	Aravalli (near Pushkar/Ajmer)	Rann of Kutch (does not reach sea)	RJ, GJ	Saline below Balotara. Only major river that ends in Rann of Kutch — not true drainage to sea.
Periyar	Cardamom Hills (Sivagiri peak), Kerala	Arabian Sea (Vembanad Lake)	Kerala	Longest river of Kerala. Idukki Dam on it. Periyar Wildlife Sanctuary (elephants).

NOTE: Narmada and Tapi are the only two major peninsular rivers that flow through rift valleys — this is a frequently tested distinction. All others flow over plateau surfaces.

PENINSULAR RIVERS — EAST FLOWING (DRAIN INTO BAY OF BENGAL)

River	Origin	Drains Into	States	Key Feature
Godavari	Trimbakeshwar, Nashik (Maharashtra)	Bay of Bengal	MH, TS, AP	"Dakshin Ganga" (Ganga of the South) — largest peninsular river. Tributaries: Pranhita, Indravati, Sabari. Papikonda Gorge (AP).

Krishna	Mahabaleshwar (Maharashtra)	Bay of Bengal	MH, KA, TS, AP	Major tributaries: Bhima, Tungabhadra, Koyna, Musi. Nagarjunasagar Dam on it.
Kaveri (Cauvery)	Brahmagiri hills, Kodagu (Karnataka)	Bay of Bengal	KA, TN	Tributaries: Hemavati, Kabini, Bhavani, Amaravati. Ranganathittu (bird sanctuary) on it. Subject of Cauvery Water Disputes Tribunal (1990).
Mahanadi	Sihawa, Dhamtari (Chhattisgarh)	Bay of Bengal (Odisha)	CG, Odisha	Hirakud Dam (Odisha) — longest earthen dam in India. Mahanadi Water Disputes Tribunal (2018).
Damodar	Chotanagpur plateau (Jharkhand)	Joins Hooghly (Ganga distributary)	JH, WB	"Sorrow of Bengal" — historically flooded West Bengal. Damodar Valley Corporation (DVC) — India's first multi-purpose river project (modelled on TVA, USA).
Tungabhadra	Confluence of Tunga + Bhadra rivers (Shivamogga area, Karnataka)	Joins Krishna (AP)	KA, AP	Hampi (Vijayanagara capital) on its banks. Tungabhadra Dam (1953).
Chambal	Vindhya Range, Mhow (Indore, MP)	Joins Yamuna (UP)	MP, RJ, UP	National Chambal Sanctuary (gharials, gangetic dolphins). Known for ravines in MP/Rajasthan. Gandhi Sagar Dam.

NOTE: Damodar joins Hooghly (not Ganga directly). Chambal and Betwa join Yamuna — not Ganga. UPSC has used these as wrong options in matching pair questions.

INTER-STATE RIVER WATER DISPUTES TRIBUNALS

Under the Inter-State River Water Disputes Act, 1956 (Parliament cannot legislate on this — it must refer to a Tribunal under Article 262).

Tribunal	Year	River(s)	States	Award / Status
Godavari WDT	1969	Godavari	AP (now TS+AP), MP, KA, MH, Odisha	Award given 1980
Krishna WDT I	1969	Krishna	MH, KA, AP	Award 1973; reconstituted 2004
Narmada WDT	1969	Narmada	MP, MH, GJ, RJ	Award 1979 (Sardar Sarovar height linked to award)
Ravi-Beas	1986	Ravi, Beas	Punjab, Haryana, Rajasthan	Linked to SYL (Sutlej-Yamuna Link) canal dispute — unresolved

Cauvery WDT	1990	Cauvery	KA, TN, Kerala, Puducherry	Final award 2007; SC modified 2018
Mahadayi (Mhadei) WDT	2010	Mahadayi (Mandovi)	Goa, MH, KA	Award August 2023; Goa vs KA diversion dispute
Vansadhara WDT	2010	Vansadhara	AP, Odisha	Award pending
Mahanadi WDT	2018	Mahanadi	Odisha, Chhattisgarh	Most recently constituted; Odisha vs CG dam construction

NOTE: Article 262 empowers Parliament to exclude Supreme Court jurisdiction over inter-state water disputes. The Act of 1956 operationalises this. Tribunal awards are final and binding like SC decrees.

QUICK REFERENCE — UNIQUE FEATURES & EXAM FACTS

Feature / Tag	River	Exact Fact (Exam-Ready)
"Sorrow of Bihar"	Kosi	Shifts course frequently (over 120 km in historical record); catastrophic floods in north Bihar
"Sorrow of Bengal"	Damodar	Flooded West Bengal plains repeatedly before DVC dams; DVC was India's first multi-purpose project
"Dakshin Ganga"	Godavari	Longest peninsular river (~1465 km); has largest drainage basin among peninsular rivers
Rift valley rivers	Narmada, Tapi	Flow through tectonic graben (down-faulted blocks) — not over plateau surface. Both form estuaries, not deltas.
National River	Ganga	Declared India's National River in 2008; largest basin area (~860,000 sq km); longest river in India (~2,525 km)
Crosses Tropic of Cancer twice	Mahi	Unique geographical characteristic; flows through Rajasthan and Gujarat
Does not reach the sea	Luni	Ends in Rann of Kutch; becomes saline below Balotara (Rajasthan)
Longest tributay of Ganga	Yamuna	~1376 km; meets Ganga at Prayagraj at Triveni Sangam (third: Saraswati — mythical)
Indus Waters Treaty	Indus system	1960, India-Pakistan, World Bank mediated. India: Ravi, Beas, Sutlej (eastern). Pakistan: Indus, Jhelum, Chenab (western).
Tehri Dam	Bhagirathi (Ganga)	Highest dam in India (260m); Uttarakhand; Bhagirathi + Alaknanda = Ganga at Devprayag
Hirakud Dam	Mahanadi	Sambalpur, Odisha; longest earthen dam in India (~26 km)

Bhakra-Nangal Dam	Sutlej	Himachal Pradesh; second-highest gravity dam in Asia; Gobind Sagar reservoir
Hampi (UNESCO heritage)	Tungabhadra	Vijayanagara Empire capital; banks of Tungabhadra, Karnataka

RECENT IN NEWS (2023–2025)

Item	River / Region	Why UPSC-Relevant
Ken-Betwa River Link Project	Ken, Betwa (MP)	First project under National River Linking Project (NRLP). Cabinet approved Dec 2021; construction phase 2024. Panna Tiger Reserve (Ken flows through it) — environment clearance controversy.
Mahadayi Tribunal Award	Mahadayi (Mandovi), Goa	Award August 2023 — Karnataka allowed to divert 13.42 TMC water; Goa contested this. Tests knowledge of tribunal finality and Article 262.
Cauvery dispute — SC 2023	Cauvery (KA-TN)	SC directed Cauvery Water Management Authority to implement 2018 modified award. Karnataka seeking review citing groundwater depletion. Recurrent prelims topic.
Namami Gange — Phase II	Ganga	Ongoing NMCG (National Mission for Clean Ganga) programme; key: Ganga is managed by MoJS (Jal Shakti), not MoEFCC. Sewage treatment plants in 97 towns.
Pancheshwar Multipurpose Project	Mahakali River (India-Nepal border)	India-Nepal bilateral project. Mahakali River = Sharda (in India). Tests trans-boundary river knowledge.
Indus Waters Treaty — India's notice	Indus (Pakistan-India)	India issued notice to Pakistan in 2023 to modify IWT (first time since 1960). Tested as current affairs + treaty knowledge combination.

★ PYQ Insight:

2024 — Asked the west-to-east order of Himalayan rivers joining Ganga downstream of Prayagraj. Answer: Gomti, Ghaghra, Gandak, Kosi (left-bank tributaries in west-to-east sequence). This is the most specific river geography question in the 15-year dataset.

2022–23 — Matching pairs: rivers with their tributaries. Common trap: students confuse Tungabhadra (joins Krishna) with Son (joins Ganga). Also tested: rivers flowing through which states.

2019–20 — Statement-based questions on Brahmaputra: "Brahmaputra originates in India" (FALSE — originates in Tibet). "Brahmaputra is a trans-boundary river" (TRUE). Both statements in one question — Statement I/II format.

Recurring trap — Left vs. right bank tributaries. Left bank = river's left side when facing downstream. Ganga left-bank: Ramganga, Gomti, Ghaghra, Gandak, Kosi. Right-bank: Yamuna, Son, Damodar. UPSC exploits this confusion every 2–3 years.

3 THINGS UPSC TESTS FROM THIS TOPIC

- **East vs. West flowing peninsular rivers.** Most peninsular rivers drain east into the Bay of Bengal. Only Narmada, Tapi, Sabarmati, Mahi, and Periyar flow west into the Arabian Sea. UPSC asks which of the given rivers "flows westward" — Narmada and Tapi are the two most tested. The additional fact: both flow in rift valleys, which makes them geologically distinct from plateau-draining rivers.
- **River-specific unique features.** "Sorrow of Bengal" = Damodar. "Sorrow of Bihar" = Kosi. "Dakshin Ganga" = Godavari. Narmada and Tapi form estuaries (not deltas). Mahi crosses the Tropic of Cancer twice. Luni becomes saline below Balotara. Brahmaputra has multiple names across countries. These appear as Statement I/II questions — knowing the why behind each fact matters.
- **Inter-state river tribunals.** UPSC tests: which tribunal, which river, which states, and what year. Cauvery (1990), Mahadayi/Mhadei (2010), Mahanadi (2018) are the most recent and most tested. Common trap: students attribute the wrong year or the wrong states. Note that all tribunals are under the Inter-State River Water Disputes Act, 1956.

BEYOND PREPARATION

Neti Academy is more than current affairs.

Mental Health Support — You are not alone. We connect aspirants with professionals who understand the pressure of UPSC preparation.

Plan B Support — If you are reconsidering your path, we help you think clearly about your options — without judgment.

Mentorship — Direct access to experienced mentors who have been through the process and can guide you at every stage.

For any of the above, write to us: admin@netiacademy.com

CONNECT WITH US

Website: www.netiacademy.com

Telegram: t.me/netiacademy01

Mentorship, Mental Health & Plan B: admin@netiacademy.com

Found an error or want to share feedback? Write to us at feedback@netiacademy.com

© 2026 Neti Academy. All rights reserved.

This document is the intellectual property of Neti Academy. Reproduction, redistribution, or resale in any form without prior written permission is strictly prohibited.

Content is curated from publicly available sources for educational purposes only.